in place of sodium light, how many fringes will be seen?

[Ans: 84]

3. Light of wavelength 600 nm that falls on a pair of slits producing interference pattern on a screen in which the bright fringes are separated by 7.2 mm. What must be the wavelength of another light which produces bright fringes separated by 8.1 mm with the same apparatus?

[Ans: 675 nm]

4. A beam of light of wavelength 600 nm from a distant source falls on a single slit 1 mm wide and the resulting diffraction pattern is observed on a screen 2 m away. What is the distance between the first dark fringes on either side of the central bright fringe?

[Ans: 2.4 mm]

5. Light of wavelength of 5000 Å produces diffraction pattern of the single slit of width 2.5 μm. What is the maximum order of diffraction possible?

[Ans: 5]

6. I_0 is the intensity of light existing between two cross Polaroids kept with their axes perpendicular to each other. A third polaroid is introduced between them. What must be the angle between the axes of first and the newly introduced polaroid to get the maximum light from the whole arrangement?

[Ans: 45°]

7. An unpolarised light of intensity 32 Wm^{-2} passes through three Polaroids such that the axes of the first and the last Polaroids are at 90°. What is the angle between the axes of the first and middle Polaroids so that the emerging light has an intensity of only 3 Wm^{-2}?

[Ans: 30°]

8. The reflected light is found to be plane polarised when an unpolarized light falls on a denser medium at 60° with the normal. Find the angle of refraction and critical angle of incidence for total internal reflection in the denser to rarer medium reflection.

[Ans: 30°, 35.15°]

9. The near point and the far point for a person are 50 cm and 500 cm, respectively. Calculate the power of the lens the person should wear to read a book held in hand at 25 cm. What maximum distance is clearly visible for the person with this lens on the eye?

[Ans: 2D, 45.45 cm]

10. A compound microscope has a magnifying power of 100 when the image is formed at infinity. The objective has a focal length of 0.5 cm and the tube length is 6.5 cm. What is the focal length of the eyepiece.

[Ans: 2 cm]

UNIT-7(XII-Physics_Vol-2)_new.indd 105 21-07-2020 16:36:45